Sevoflurane: Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Sevoflurane: Patient drug information”

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Ultane

Pharmacologic Category

- General Anesthetic, Inhalation

Dosing: Adult

Anesthesia

Anesthesia: Inhalation: Surgical levels of anesthesia are generally achieved with concentrations from 0.5% to 3% with or without the concomitant use of nitrous oxide; the concentration at which amnesia and loss of awareness occur is 0.6% (Ref).

MAC values for surgical levels of anesthesia:

25 years:

Sevoflurane in oxygen: 2.6%

Sevoflurane in 65% N_20/35% oxygen: 1.4%

40 years:

Sevoflurane in oxygen: 2.1%

Sevoflurane in 65% N_20/35% oxygen: 1.1%

60 years:

Sevoflurane in oxygen: 1.7%

Sevoflurane in 65% N_20/35% oxygen: 0.9%

80 years:

Sevoflurane in oxygen: 1.4%

Sevoflurane in 65% N_20/35% oxygen: 0.7%

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. However, need for dosage adjustment unlikely as limited excretion in the urine.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in manufacturer's labeling; use with caution (safety with severe hepatic impairment has not been studied).

Dosing: Older Adult

Refer to adult dosing. MAC is reduced in the elderly (50% reduction by age 80).

Dosing: Pediatric

Anesthesia

Anesthesia: Inhalation: Surgical levels of anesthesia are generally achieved with concentrations from 0.5% to 3% with or without the concomitant use of nitrous oxide; the concentration at which amnesia and loss of awareness occur is 0.6% (Ref).

MAC values for surgical levels of anesthesia:

0 to 1 month old full-term neonates: Sevoflurane in oxygen: 3.3%

1 to <6 months: Sevoflurane in oxygen: 3%

6 months to <1 year:

Sevoflurane in oxygen: 2.8%

Sevoflurane in 65% N_20/35% oxygen: 2%

1 to <3 years:

Sevoflurane in oxygen: 2.8%

Sevoflurane in 60% N_20/40% oxygen: 2%

3 to 12 years: Sevoflurane in oxygen: 2.5%

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling; use with caution; has not been studied in adult patients with elevated SCr.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in manufacturer's labeling; use with caution (safety with severe hepatic impairment has not been studied).

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported in adult and pediatric patients.

>10%:

Cardiovascular: Hypotension (4% to 11%)

Gastrointestinal: Nausea (25%), vomiting (18%)

Nervous system: Agitation (7% to 15%)

Respiratory: Increased cough (5% to 11%)

1% to 10%:

Cardiovascular: Bradycardia (5%), hypertension (2%), tachycardia (2% to 6%)

Gastrointestinal: Sialorrhea (2% to 4%)

Nervous system: Dizziness (4%), drowsiness (9%), headache (1%), hypothermia (1%), myoclonus (1%), shivering (6%)

Neuromuscular & skeletal: Laryngospasm (2% to 8%)

Respiratory: Airway obstruction (8%), apnea (2%), breath-holding (2% to 5%)

Miscellaneous: Fever (1%)

<1%:

Cardiovascular: Atrial arrhythmia, atrial fibrillation, bigeminy, cardiac arrhythmia, complete atrioventricular block, depression of ST segment on ECG, inversion T wave on ECG, premature ventricular contractions, second-degree atrioventricular block, supraventricular extrasystole, syncope

Dermatologic: Pruritus, skin rash

Endocrine & metabolic: Acidosis, albuminuria, hyperglycemia, hypophosphatemia, increased lactate dehydrogenase

Gastrointestinal: Dental fluorosis, dysgeusia, hiccups, xerostomia

Genitourinary: Glycosuria, oliguria, urinary retention, urination disorder, urine abnormality

Hematologic & oncologic: Hemorrhage, leukocytosis, thrombocytopenia

Hepatic: Hyperbilirubinemia, increased serum alanine aminotransferase, increased serum alkaline phosphatase, increased serum aspartate aminotransferase

Nervous system: Asthenia, confusion, crying, hypertonia, insomnia, nervousness, pain

Ophthalmic: Amblyopia, conjunctivitis

Renal: Increased blood urea nitrogen, increased serum creatinine

Respiratory: Bronchospasm, dyspnea, hyperventilation, hypoventilation, hypoxia, increased bronchial secretions, pharyngitis, stridor, wheezing

Miscellaneous: Malignant hyperthermia

Postmarketing:

Cardiovascular: Chest discomfort, prolonged QT interval on ECG, torsades de pointes

Hepatic: Hepatic failure, hepatic necrosis, hepatitis, hepatotoxicity (idiosyncratic; Chalasani 2021), jaundice

Hypersensitivity: Anaphylaxis, hypersensitivity reaction

Nervous system: Delirium, seizure

Respiratory: Pulmonary alveolar hemorrhage

Contraindications

Hypersensitivity to sevoflurane, other halogenated inhalational anesthetics, or any component of the formulation; known or suspected genetic susceptibility to malignant hyperthermia.

Canadian labeling: Additional contraindications (not in US labeling): Occurrence of liver dysfunction, jaundice or unexplained fever, leukocytosis, or eosinophilia after previous halogenated anesthetic administration; when general anesthesia is contraindicated.

Warnings/Precautions

Concerns related to adverse effects:

• Agitation/delirium: Monitor for emergence agitation or delirium (Stachnik 2006).

• CNS depression: May cause CNS depression, which may impair physical or mental abilities; patients must be cautioned about performing tasks that require mental alertness (eg, operating machinery, driving).

• Hepatic effects: Postoperative hepatitis (eg, jaundice associated with fever and/or eosinophilia) or hepatic dysfunction has been reported; prior exposure to halogenated hydrocarbon anesthetics may increase this risk.

• Hyperkalemia: Use of inhaled anesthetics has been associated with rare cases of perioperative hyperkalemia in pediatric patients; concomitant use of succinylcholine was associated with most of the reported cases, but not all. Patients with latent and overt neuromuscular disease (eg, Duchenne muscular dystrophy) are the most vulnerable. Other abnormalities may include elevation in CPK and myoglobinuria. Monitor closely for arrhythmias. Aggressively identify and treat hyperkalemia and resistant arrhythmias.

• Hypotension: Sevoflurane produces a dose-dependent reduction in blood pressure during maintenance of anesthesia and may occur more rapidly compared to other inhaled anesthetics.

• Increased intracranial pressure: May dilate the cerebral vasculature and may, in certain conditions, increase intracranial pressure (Stachnik 2006).

• Malignant hyperthermia: May trigger malignant hyperthermia; some reported cases have been fatal. Risk may be increased with concomitant administration of succinylcholine and volatile anesthetic agents and patients with genetic factors or family history of malignant hyperthermia, including ryanodine receptor or dihydropyridine receptor inherited variants. Signs of malignant hyperthermia may include arrhythmias, cyanosis, hemodynamic instability, hypercapnia, hyperthermia, hypovolemia, hypoxia, muscle rigidity, tachycardia, and tachypnea; coagulopathies, renal failure, and skin mottling may also occur. If malignant hyperthermia is suspected, discontinue triggering agents and institute appropriate therapy (eg, dantrolene) and other supportive measures.

• QT prolongation: Cases of QT prolongation in association with torsade de pointes (some fatal) have been reported with sevoflurane use; use caution when administering to patients at risk of QT prolongation (eg, concurrent use of drugs that can prolong the QT interval such as class Ia and III antiarrhythmic drugs, elderly patients, congenital QT prolongation) (Han 2010; Kang 2006; Nakao 2010).

• Respiratory depression: May cause dose-dependent respiratory depression and blunted ventilatory response to hypoxia and hypercapnia (Golembiewski 2004); response may be augmented by other medications with respiratory depressant effect (eg, opioids). Hypoxic pulmonary vasoconstriction is blunted, which may lead to increased pulmonary shunt (Miller 2010).

Disease-related concerns:

• Heart failure: In a scientific statement from the American Heart Association, sevoflurane has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: major) (AHA [Page 2016]).

• Hepatic impairment: Use with caution in patients with hepatic impairment; safety with severe impairment has not been established.

• Renal impairment: Use with caution in patients with renal impairment (ie, creatinine >1.5 mg/dL); safety with severe impairment has not been established.

• Seizure disorder: Use with caution in patients at risk for seizures; seizures have been reported in children and young adults.

Special populations:

• Pediatric neurotoxicity: In pediatric and neonatal patients <3 years and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems, including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Special handling:

• Occupational caution: There is no specific work exposure limit established for sevoflurane. However, the National Institute for Occupational Safety and Health (NIOSH) has recommended an 8 hour time-weighted average limit of 2 ppm for halogenated anesthetic agents in general (0.5 ppm when coupled with exposure to N_2O).

Other warnings/precautions:

• Desiccated absorbents: Reaction of sevoflurane with CO_2 absorbents that become desiccated within circle breathing equipment can lead to formation of formaldehyde (causing respiratory irritation) and carbon monoxide; maintain fresh absorbent as per manufacturer guidelines regardless of state of colorimetric indicator. An exothermic reaction occurs when sevoflurane is exposed to CO_2 absorbents; this reaction is increased when the CO_2 absorbent becomes desiccated. Rare cases of extreme heat, smoke, and/or fire within breathing circuit have been reported. This reaction also leads to formation of a fluorinated byproduct, compound A, which has been reported to cause mild and reversible renal injury in animal studies (Gentz 2001). The theoretical risk of compound A-induced nephrotoxicity in humans may be dose- and exposure time-dependent; minimize exposure risk by not exceeding 2 MAC hours and using fresh flow rates of 1 to <2 L/minute (low fresh gas flow rates maximize rebreathing of the anesthetic). Steps to help reduce the risk of these events (eg, shutting off the anesthesia machine at the end of clinical use or after any case when a subsequent extended period of nonuse is expected, routine replacement of the CO_2 absorbents) should be incorporated into routine practice (Miller 2010).

Warnings: Additional Pediatric Considerations

Pediatric patients with trisomy 21 (Down syndrome) have been shown to be at an increased risk of developing bradycardia during anesthetic induction with sevoflurane. The mechanism for bradycardia is unknown. Some patients have failed to respond to traditional treatment options (eg, atropine). Dose reduction of sevoflurane and adequate oxygenation and ventilation were effective in most cases to reestablish baseline heart rates and hemodynamic stability (Bai 2010; Health Canada Safety Review 2018; Kraemer 2010; Roodman 2003; Walia 2016).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Inhalation:

Ultane: (250 mL)

Generic: (250 mL)

Generic Equivalent Available: US

Yes

Administration: Adult

Via sevoflurane-specific calibrated vaporizers; use cautiously in low-flow or closed-circuit systems since sevoflurane is unstable and potentially toxic breakdown products have been liberated.

Administration: Pediatric

Inhalation: Administer via sevoflurane-specific calibrated vaporizers; use cautiously in low-flow or closed-circuit systems since sevoflurane is unstable and potentially toxic breakdown products have been liberated.

Storage/Stability

Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Anesthesia: Induction and maintenance of general anesthesia in adults and pediatric patients for inpatient and outpatient surgery

Medication Safety Issues

Sound-alike/look-alike issues:

Ultane may be confused with Ultram

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

Substrate of CYP2A6 (Minor), CYP2B6 (Minor), CYP2E1 (Minor), CYP3A4 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acrivastine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amphetamines: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Bambuterol: May increase arrhythmogenic effects of Inhalational Anesthetics. Management: Some labels recommend specifically avoiding halothane; others recommend separating administration by at least 6 hours; other bambuterol labels do not mention this possible interaction. Monitor for increased sensitivity to arrhythmias if coadministered. *Risk D: Consider Therapy Modification*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Calcium Channel Blockers: Inhalational Anesthetics may increase hypotensive effects of Calcium Channel Blockers. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dabrafenib: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

DexmedeTOMIDine: CNS Depressants may increase CNS depressant effects of DexmedeTOMIDine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Dexmethylphenidate-Methylphenidate: May increase hypertensive effects of Inhalational Anesthetics. Management: Avoid the use of dexmethylphenidate/methylphenidate on the day of a planned surgery with an inhalational anesthetic. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimenhydrinate: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

DOPamine: Inhalational Anesthetics may increase arrhythmogenic effects of DOPamine. *Risk C: Monitor*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxapram: May increase adverse/toxic effects of Inhalational Anesthetics. Management: Delay administration of doxapram until the inhalational anesthetic has been excreted (ie, after 5 half-lives) in order to lessen the potential for arrhythmias, including ventricular tachycardia and ventricular fibrillation. *Risk D: Consider Therapy Modification*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

DroPERidol: May increase CNS depressant effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). DroPERidol may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Consider dose reductions and monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ephedra: May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk C: Monitor*

EPHEDrine (Nasal): May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk X: Avoid*

EPINEPHrine (Systemic): Inhalational Anesthetics may increase arrhythmogenic effects of EPINEPHrine (Systemic). Management: Administer epinephrine with added caution in patients receiving, or who have recently received, inhalational anesthetics. Use lower than normal doses of epinephrine and monitor for the development of cardiac arrhythmias. *Risk D: Consider Therapy Modification*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Injection): May increase sedative effects of Inhalational Anesthetics. *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Fenfluramine: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Fenoterol: Inhalational Anesthetics may increase arrhythmogenic effects of Fenoterol. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fluorouracil Products: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Haloperidol: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

HydrOXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Isocarboxazid: May increase hypotensive effects of Inhalational Anesthetics. *Risk X: Avoid*

Isoproterenol: Inhalational Anesthetics may increase arrhythmogenic effects of Isoproterenol. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaraminol: Inhalational Anesthetics may increase arrhythmogenic effects of Metaraminol. *Risk X: Avoid*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methadone: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase CNS depressant effects of Methadone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation, ventricular arrhythmias, sedation, and respiratory depression. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: Sevoflurane may increase nephrotoxic effects of Methoxyflurane. *Risk X: Avoid*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Monoamine Oxidase Inhibitors: May increase adverse/toxic effects of Sevoflurane. Specifically, the risk of hemodynamic instability may be increased. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Inhalational Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Management: When initiating a non-depolarizing neuromuscular blocking agent (NMBA) in a patient receiving an inhalational anesthetic, initial NMBA doses should be reduced 15% to 25% and doses of continuous infusions should be reduced 30% to 60%. *Risk D: Consider Therapy Modification*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Norepinephrine: Inhalational Anesthetics may increase arrhythmogenic effects of Norepinephrine. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ondansetron: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pentamidine (Systemic): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenylephrine (Ophthalmic): May increase adverse/toxic effects of Inhalational Anesthetics. Specifically, the cardiovascular depressant effects of inhalational anesthetics may be increased. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Inhalational Anesthetics. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antidepressants (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): May increase hypotensive effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for hypotension and QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reproterol: Inhalational Anesthetics may increase arrhythmogenic effects of Reproterol. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ritodrine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Sevoflurane crosses the placenta (Satoh 1995).

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D- aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity, may affect brain development. Evaluate benefits and potential risks of fetal exposure to sevoflurane when duration of surgery is expected to be >3 hours (Olutoye 2018).

Use of sevoflurane in obstetric anesthesia has been described (ACOG 209 2019; Choi 2012; Devroe 2015; Gambling 1995; Karaman 2006). Maternal exposure should be minimized due to dose dependent uterine relaxation and fetal depression (ACOG 209 2019; Devroe 2015). Adverse events have not been observed following use as part of general anesthesia for elective cesarean delivery.

The ACOG recommends that pregnant women should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

It is not known if sevoflurane is present in breast milk.

Patients can express and discard milk for the first 24 hours after administration to minimize infant exposure via breast milk.

The Academy of Breastfeeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

BP, temperature, heart rate and rhythm, respiration, oxygen saturation, end-tidal CO_2 and end-tidal sevoflurane concentrations should be monitored prior to and throughout anesthesia; temperature of CO_2 absorbent canister

Mechanism of Action

Inhaled anesthetics alter activity of neuronal ion channels particularly the fast synaptic neurotransmitter receptors (nicotinic acetylcholine, GABA, and glutamate receptors). Limited effects on sympathetic stimulation including cardiovascular system. Sevoflurane does not cause respiratory irritation or circulatory stimulation. May depress myocardial contractility, decrease blood pressure through a decrease in systemic vascular resistance and decrease sympathetic nervous activity.

Pharmacokinetics (Adult Data Unless Noted)

Sevoflurane has a low blood/gas partition coefficient and therefore is associated with a rapid onset of anesthesia and recovery

Onset of action: Time to induction: Within 2 to 3 minutes

Duration: Emergence time: Depends on blood concentration when sevoflurane is discontinued. The rate of change of anesthetic concentration in the lung is rapid with sevoflurane because of its low blood gas solubility (0.63). The 90% decrement time (time required for anesthetic concentration in vessel-rich tissues to decrease by 90%) for sevoflurane is short when the duration of anesthesia is <2 hours but increases dramatically as the duration of administration is lengthened (Bailey, 1997).

Metabolism: Hepatic (~5%) via CYP2E1

Excretion: Exhaled gases

Patient Counseling Points

(For additional information see "Sevoflurane: Patient drug information")

What is this drug used for?

• It is used for anesthesia during some procedures or surgery.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Feeling sleepy

• Upset stomach or throwing up

• Shivering

• Cough

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain

• High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath

• Liver problems like dark urine, feeling tired, not hungry, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes

• Very bad dizziness or passing out

• Trouble breathing, slow breathing, or shallow breathing

• Slow heartbeat

• Seizures

• Feeling agitated

• Muscle stiffness

• Change in color of skin to a bluish color like on the lips, nail beds, fingers, or toes

• Malignant hyperthermia like a fast heartbeat, fast breathing, fever, or spasm or stiffness of the jaw muscles

• Abnormal heartbeat (prolonged QT interval) like a fast or abnormal heartbeat, or if you pass out

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Eraldin | Sevocris | Sevocristalia | Sevoflurano baxter | Sevomax | Sevorane;
- (AT) Austria: Sevofluran baxter | Sevofluran piramal | Sevofluran zeosys | Sevorane;
- (AU) Australia: Sevoflurane sandoz | Sevorane;
- (BD) Bangladesh: Sevorane;
- (BE) Belgium: Sevoflurane baxter | Sevorane;
- (BG) Bulgaria: Sevorane;
- (BR) Brazil: Anesevo | Sevocris | Sevoflurano | Sevoness | Sevoq | Sevorane | Voflur;
- (CH) Switzerland: Sevoflurane baxter;
- (CL) Chile: Sevoflurano | Sevorane | Sojourn;
- (CN) China: Kai te li | Qi fu mei | Qifumei | Yi jun ning;
- (CO) Colombia: Bloktec | Sevoflurano | Sevorane | Sojourn;
- (CR) Costa Rica: Floves | Sevoflurano | Sevorane | Sevotroy | Sojourn;
- (CZ) Czech Republic: Sevorane | Sojourn;
- (DE) Germany: Sevorane;
- (DK) Denmark: Sevorane;
- (DO) Dominican Republic: Sevocris | Sevorane;
- (EC) Ecuador: Inharane | Seflur | Sevocris | Sevoflurano | Sevorane | Sevotroy | Sojourn;
- (EE) Estonia: Sevoflurane baxter | Sevoflurane piramal | Sevorane;
- (EG) Egypt: Sevoflurane usp;
- (ES) Spain: Sevoflurano | Sevoflurano baxter | Sevoflurano piramal | Sevorane;
- (FI) Finland: Sevorane;
- (FR) France: Sevoflurane baxter | Sevorane | Sojourn;
- (GB) United Kingdom: Sevoflurane abbott | Sevorane;
- (GR) Greece: Sevorane;
- (HK) Hong Kong: Sevorane;
- (HR) Croatia: Sevofluran baxter | Sevorane;
- (HU) Hungary: Sevoflurane baxter | Sevoflurane piramal | Sevorane;

- (ID) Indonesia: Sevodex | Sevorane | Sojourn;
- (IE) Ireland: Sevoflurane baxter | Sevorane;
- (IN) India: Sevfurane | Sevitrue | Sevorane | Sevotroy | Sevura | Sojourn;
- (IT) Italy: Sevoflurane baxter | Sevoflurane piramal | Sevorane;
- (JP) Japan: Sevofrane abbott | Sevofrane aventis | Sevofrane maruishi | Sevoness;
- (KE) Kenya: Sevotroy | Ultane;
- (KR) Korea, Republic of: Baxter sevoflurane | Sevofran | Sevofrane | Sevoprane | Sevorane | Sojourn;
- (LT) Lithuania: Sevoflurane baxter | Sevoflurane piramal | Sevorane;
- (LV) Latvia: Sevoflurane piramal | Sevoran | Sevorane;
- (MA) Morocco: Sevorane;
- (MX) Mexico: Fhadotil | Floves | Savannlab | Sevam | Sevoflurano | Sevorane | Sovener | Svofast | Zurenpharm;
- (MY) Malaysia: Sevorane | Sojourn;
- (NG) Nigeria: Ultane;
- (NL) Netherlands: Sevoflurane baxter | Sevorane;
- (NO) Norway: Servofluran Baxter | Sevorane;
- (NZ) New Zealand: Sevorane;
- (PA) Panama: Floves | Sevoflurano baxter | Sevorane | Sojourn;
- (PE) Peru: Inharane | Sevoflurano | Sevogesic | Sevorane | Sojourn;
- (PH) Philippines: Baxter sevoflurane | Sevo | Sevorane | Sojourn;
- (PK) Pakistan: Sevorane | Sojourn;
- (PL) Poland: Sevoflurane baxter | Sevorane | Sojourn;
- (PR) Puerto Rico: Sojourn | Ultane;
- (PT) Portugal: Sevoflurano | Sevoflurano ojourn | Sevorane;
- (PY) Paraguay: Eraldin | Floves | Sevocris | Sevoflurano omicron | Sevoflurano prosalud | Sevoflurano quimfa | Sevoflurano usp interlabo | Sevorane;
- (QA) Qatar: Sojourn;
- (RO) Romania: Sevo anesteran;
- (RU) Russian Federation: Sevakton | Sevofluran medisorb | Sevofluran vial | Sevoflurane vial | Sevorane | Sojurne;
- (SA) Saudi Arabia: Haluran | Sojourn;
- (SE) Sweden: Sevofluran baxter | Sevorane;
- (SG) Singapore: Sevorane | Sojourn;
- (SI) Slovenia: Sevofluran baxter | Sevofluran piramal | Sevorane;
- (SK) Slovakia: Sevofluran baxter | Sevorane | Sojourn;
- (SV) El Salvador: Floves;
- (TH) Thailand: Sefether | Sevoflurane baxter | Sevorane | Sojourn;
- (TN) Tunisia: Sevoflurane usp | Sevorane;
- (TR) Turkey: Sevoflurane baxter | Sevones | Sevorane | Sojourn;
- (TW) Taiwan: Sevoflurane baxter | Sevofrane | Sojourn | Ultane;
- (UA) Ukraine: Hipnoranum | Sevo anesteran | Sevorane;
- (UG) Uganda: Sevotroy;
- (UY) Uruguay: Sevocris | Sevorane | Sojourn;
- (VE) Venezuela, Bolivarian Republic of: Sevorane | Sojourn;
- (ZA) South Africa: Sojourn | Ultane;
- (ZM) Zambia: Ultane;
- (ZW) Zimbabwe: Ultane

REFERENCES

1. American College of Obstetricians and Gynecologists (ACOG). ACOG practice bulletin no. 209: obstetric analgesia and anesthesia. *Obstet Gynecol*. 2019;133(3):e208-e225. [PubMed 30801474]

2. American College of Obstetricians and Gynecologists (ACOG). ACOG committee opinion no. 775: nonobstetric surgery during pregnancy. *Obstet Gynecol*. 2019;133(4):e285-e286. [PubMed 30913200]

3. Bai W, Voepel-Lewis T, Malviya S. Hemodynamic changes in children with Down syndrome during and following inhalation induction of anesthesia with sevoflurane. *J Clin Anesth*. 2010;22(8):592-597. [PubMed 21109130]

4. Bailey JM. Context-sensitive half-times and other decrement times of inhaled anesthetics. *Anesth Analg*. 1997;85(3):681-686. [PubMed 9296431]

5. Beck-Schimmer B, Breitenstein S, Urech S, et al, “A Randomized Controlled Trial on Pharmacological Preconditioning in Liver Surgery Using a Volatile Anesthetic,” *Ann Surg*, 2008, 248(6):909-18. [PubMed 19092335]

6. Bein B, Renner J, Caliebe D, et al, “The Effects of Interrupted or Continuous Administration of Sevoflurane on Preconditioning Before Cardio-Pulmonary Bypass in Coronary Artery Surgery: Comparison With Continuous Propofol,” *Anaesthesia*, 2008, 63(10):1046-55. [PubMed 18627368]

7. Campagna JA, Miller KW, and Forman SA, “Mechanisms of Action of Inhaled Anesthetics,” *N Engl J Med*, 2003, 348(21):2110-24. [PubMed 12761368]

8. Chalasani NP, Maddur H, Russo MW, Wong RJ, Reddy KR; Practice Parameters Committee of the American College of Gastroenterology. ACG Clinical Guideline: Diagnosis and Management of Idiosyncratic Drug-Induced Liver Injury. *Am J Gastroenterol*. 2021;116(5):878-898. doi:10.14309/ajg.0000000000001259 [PubMed 33929376]

9. Choi WJ, Kim SH, Koh WU, et al. Effect of pre-exposure to sevoflurane on the bispectral index in women undergoing Caesarean delivery under general anaesthesia. *Br J Anaesth*. 2012;108(6):990-997. doi: 10.1093/bja/aes036. [PubMed 22434266]

10. De Hert SG, Turani F, Mathur S, et al, “Cardioprotection With Volatile Anesthetics: Mechanisms and Clinical Implications,” *Anesth Analg*, 2005, 100(6):1584-93. [PubMed 15920178]

11. Devroe S, Van de Velde M, Rex S. General anesthesia for caesarean section. *Curr Opin Anaesthesiol*. 2015;28(3):240-246. doi: 10.1097/ACO.0000000000000185. [PubMed 25827280]

12. Doi M and Ikeda K, “Airway Irritation Produced by Volatile Anaesthetics During Brief Inhalation: Comparison of Halothane, Enflurane, Isoflurane and Sevoflurane,” *Can J Anaesth*, 1993, 40(2):122-6. [PubMed 8443850]

13. Eger EI 2nd, “Characteristics of Anesthetic Agents Used for Induction and Maintenance of General Anesthesia,” *Am J Health Syst Pharm*, 2004, 61(Suppl 4):3-10.

14. Food and Drug Administration. FDA Drug Safety Communication: FDA review results in new warnings about using general anesthetics and sedation drugs in young children and pregnant women. https://www.fda.gov/Drugs/DrugSafety/ucm532356.htm. Published December 14, 2016. Accessed May 26, 2017.

15. Frink EJ Jr, Ghantous H, Malan TP, et al, “Plasma Inorganic Fluoride With Sevoflurane Anesthesia: Correlation With Indices of Hepatic and Renal Function,” *Anesth Analg*, 1992, 74(2):231-5. [PubMed 1731542]

16. Gambling DR, Sharma SK, White PF, Van Beveren T, Bala AS, Gouldson R. Use of sevoflurane during elective cesarean birth: a comparison with isoflurane and spinal anesthesia. *Anesth Analg*. 1995;81(1):90-95. [PubMed 7598289]

17. Gentz BA, Malan TP Jr. Renal toxicity with sevoflurane: a storm in a teacup? *Drugs*. 2001;61(15):2155-2162. [PubMed 11772127]

18. Golembiewski J, “Considerations in Selecting an Inhaled Anesthetic Agent: Case Studies,” *Am J Health Syst Pharm*, 2004, 61 (Suppl 4):10-17. [PubMed 15532144]

19. Han DW, Park K, Jang SB, et al, "Modeling the Effect of Sevoflurane on Corrected QT Prolongation: A Pharmacodynamic Analysis," *Anesthesiology*, 2010, 113(4):108-11. [PubMed 20808206]

20. Healthy Canadians Recalls & Alerts: Summary Safety Review - Sevoflurane - Re-assessing the potential risk of slow heartbeat (bradycardia) in children with Down syndrome. Health Canada website. Available at: https://dhpp.hpfb-dgpsa.ca/review-documents/resource/SSR00197 Published April 9, 2018.

21. Jones RM, "Desflurane and Sevoflurane: Inhalation Anaesthetics for This Decade?" *Br J Anaesth*, 1990, 65(4):527-36. [PubMed 2248821]

22. Kang J, Reynolds WP, Chen X, et al, "Mechanisms Underlying the QT interval-prolonging Effects of Sevoflurane and Its Interactions With Other QT-prolonging Drugs," *Anesthesiology*, 2006, 104(5):1015-22. [PubMed 16645454]

23. Karaman S, Akercan F, Aldemir O, Terek MC, Yalaz M, Firat V. The maternal and neonatal effects of the volatile anaesthetic agents desflurane and sevoflurane in caesarean section: a prospective, randomized clinical study. *J Int Med Res*. 2006;34(2):183-192. [PubMed 16749414]

24. Katoh T, Ijeda K. The effects of fentanyl on sevoflurane requirements for loss of consciousness and skin incision. *Anesthesiology*. 1998;88(1):18-24. [PubMed 9447851]

25. Katoh T, Suguro Y, Nakajima R, et al, "Blood Concentrations of Sevoflurane and Isoflurane on Recovery From Anaesthesia," *Br J Anaesth*, 1992, 69(3):259-62. [PubMed 1389843]

26. Kraemer FW, Stricker PA, Gurnaney HG, et al. Bradycardia during induction of anesthesia with sevoflurane in children with Down syndrome. *Anesth Analg*. 2010;111(5):1259-1263. [PubMed 20736433]

27. Miller RD. *Miller's Anesthesia*. 7th ed. Philadelphia, PA: Churchill Livingstone; 2010.

28. Nakao S, Hatano K, Sumi C, et al, "Sevoflurane Causes Greater QTc Interval Prolongation in Elderly Patients Than in Younger Patients," *Anesth Analg*, 2010, 110(3):775-9. [PubMed 20185656]

29. Olutoye OA, Baker BW, Belfort MA, Olutoye OO. Food and Drug Administration warning on anesthesia and brain development: implications for obstetric and fetal surgery. *Am J Obstet Gynecol*. 2018;218(1):98-102. [PubMed 28888583]

30. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs That May Cause or Exacerbate Heart Failure: A Scientific Statement From the American Heart Association [published correction appears in *Circulation*. 2016;134(12):e261]. *Circulation*. 2016;134(6):e32-e69. [PubMed 27400984]

31. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506. [PubMed 29624435]

32. Roodman S, Bothwell M, Tobias JD. Bradycardia with sevoflurane induction in patients with trisomy 21. *Paediatr Anaesth*. 2003;13(6):538-540. [PubMed 12846713]

33. Satoh D, Iwatsuki N, Naito M, Sato M, Hashimoto Y. Comparison of the placental transfer of halothane, enflurane, sevoflurane, and isoflurane during cesarean section. *J Anesth*. 1995;9(3):220-223. doi: 10.1007/BF02479867. [PubMed 28921220]

34. Sevorane AF (sevoflurane) [product monograph]. St-Laurent, Quebec, Canada: AbbVie Corporation; May 2021.

35. Smith I, Ding Y, and White PF, "Comparison of Induction, Maintenance, and Recovery Characteristics of Sevoflurane-N20 and Propofol-Sevoflurane-N(2)O With Propofol-Isoflurane-N(2)O Anesthesia," *Anesth Analg*, 1992, 74(2):253-9. [PubMed 1731547]

36. Stachnik J. Inhaled anesthetic agents. *Am J Health-Syst Pharm*. 2006;63(7):623-634. [PubMed 16554286]
37. Strum DP and Eger EI 2d, “Partition Coefficients for Sevoflurane in Human Blood, Saline, and Olive Oil,” *Anesth Analg*, 1987, 66(7):654-6. [PubMed 3605675]
38. Ultane (sevoflurane) [prescribing information]. North Chicago, IL: AbbVie Inc; February 2025.
39. Walia H, Ruda J, Tobias JD. Sevoflurane and bradycardia in infants with trisomy 21: A case report and review of the literature. *Int J Pediatr Otorhinolaryngol*. 2016;80:5-7. [PubMed 26746603]
40. Wang L, Traystman RJ, and Murphy SJ, “Inhalational Anesthetics as Preconditioning Agents in Ischemic Brain,” *Curr Opin Pharmacol*, 2008, 8(1):104-10. [PubMed 17962069]
41. Yasuda N, Lockhart SH, Eger EI 2nd, et al, “Comparison of Kinetics of Sevoflurane and Isoflurane in Humans,” *Anesth Analg*, 1991, 72(3):316-24. [PubMed 1994760]

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